

Do Premium Subsidies Broaden Participation or Deepen Coverage? Evidence from Livestock Price Insurance

Abstract

Livestock Risk Protection insurance provides subsidized downside price protection for cattle producers. This paper examines how subsidy intensity affects participation, insured scale, and coverage choice. We combine USDA Risk Management Agency endorsement records with matched CME livestock put prices and estimate county and year-quarter fixed-effects instrumental-variable models. The main instrument is the normalized put premium matched by cattle market, policy effective date, maturity, and log moneyness. For feeder cattle, a 10 percent increase in subsidy intensity raises the number of endorsements by about 5 percent and increases the insured-head share at coverage levels of at least 97 percent by 1.6 to 2.0 percentage points. Estimated effects on insured head and liability are positive, although their significance depends on the matching window and sample period. Average coverage changes little. Full endorsement records show that feeder LRP activity is concentrated in major cattle-producing states. A separate drought analysis finds that exceptional drought is associated with lower participation, insured head, and liability. Coverage choices among active county-quarters remain broadly unchanged. The results identify policy counts and movement into the highest coverage tier as the clearest responses to subsidy intensity.

Keywords: Livestock Risk Protection, livestock insurance, premium subsidies, cattle price risk, coverage choice, agricultural risk management, CME options

1 Introduction

Livestock producers face substantial price risk. Cattle producers are especially exposed to rapid downside price movements because biological production cycles and marketing plans adjust more slowly than market prices. Risk management through futures, options, and insurance has therefore been a central topic in agricultural economics ([Working, 1953](#); [Ederington, 1979](#); [Anderson and Danthine, 1983](#); [Moschini and Hennessy, 2001](#)). Livestock Risk Protection insurance is a federally subsidized product that protects producers against declines in expected livestock prices. Its historically limited and uneven use raises the question of whether premium subsidies can expand access.

This paper asks whether higher LRP premium subsidies broaden participation or mainly deepen coverage among producers who already use the program. The distinction matters because these responses have different policy meanings. Increases in insured head, policy counts, or liability imply broader program use. Increases in coverage level or high-coverage share imply richer protection among participants. The agricultural insurance literature has long shown that subsidies can affect both take-up and coverage choices ([Goodwin, 1993](#); [Goodwin and Smith, 1995](#); [Glauber, 2004](#); [Smith and Glauber, 2012](#); [Yu et al., 2018](#)), but these margins have not been cleanly separated in the context of livestock price insurance.

We use USDA Risk Management Agency administrative records for feeder and fed cattle LRP. The annual source files cover reinsurance years 2005 through 2026. Policy effective dates in the available full records extend from January 2005 through July 2025. The option-matched estimation sample covers 2005 through 2024. We aggregate endorsements to the county-quarter level and examine insured head, endorsement counts, liability, average coverage, and the insured-head share at coverage levels of at least 97 percent. Subsidy intensity is measured as the subsidy amount per dollar of insured liability, following the logic of the crop insurance subsidy literature ([Yu et al., 2018](#)).

Our empirical strategy uses matched CME put option prices as an instrument for LRP subsidy intensity. This design is motivated by the institutional link between LRP and livestock derivatives markets. CME put prices measure the market cost of downside price protection, while county-level LRP demand shocks are unlikely to determine national option prices. The baseline specification includes county fixed effects, year-quarter fixed effects, and contract controls. We also examine robustness across option matching windows, cattle types, state concentration, policy-period checks, and external shocks including drought and Covid-era disruptions.

The clearest response to subsidy intensity occurs in endorsement counts and high-tier coverage. In the main feeder-cattle specification, the elasticity of endorsement counts is 0.505. The coefficient on the insured-head share at coverage levels of at least 97 percent is 0.208. Both estimates are highly significant and remain stable under the stricter

matching rule. Insured head and liability also have positive coefficients. Their statistical significance varies across matching windows and sample periods. Average coverage does not respond consistently. Endorsement-level tier distributions show a large movement into the highest coverage category, and spatial analysis documents continued concentration in major cattle-producing states.

The drought evidence describes a separate source of demand variation. In a balanced county-quarter panel, exceptional drought is associated with lower feeder-cattle participation, insured head, policy counts, and liability. Conditional coverage outcomes change little among active county-quarters. These estimates indicate that production stress operates through participation and insured exposure. The drought-onset event study shows movement before first D4 classification, so we interpret the estimates as conditional associations and timing evidence.

This paper makes three contributions. It separates changes in transaction frequency, insured scale, and coverage composition using administrative endorsements. It connects public livestock insurance demand to private derivatives pricing through corrected maturity and moneyness matches. It also documents how exceptional drought changes participation and insured exposure. The evidence shows that subsidies increase LRP transactions and encourage selection of the highest coverage tier. Effects on total insured scale are less precisely determined.

The rest of the paper proceeds as follows. Section 2 describes the institutional background and related literature. Section 3 describes the data and option matching procedure. Section 4 presents the empirical strategy. Section 5 reports the main results.

2 Background and Institutional Setting

This section provides the institutional and conceptual background for the empirical design. The first subsection describes LRP as subsidized downside price insurance. The second explains why the distinction between participation and coverage choice is central for evaluating subsidy policy. The third discusses the connection between LRP and CME futures and options markets. The fourth describes drought and Covid-era disruptions as external state-contingent factors that may affect insurance demand.

2.1 LRP as Subsidized Downside Price Insurance

Livestock Risk Protection is administered by the USDA Risk Management Agency and provides protection against declines in expected livestock prices. For feeder and fed cattle, producers choose an endorsement period, number of head, target weight, and coverage level. The contract pays an indemnity when the ending value falls below the coverage

price. It provides downside price protection and does not cover physical production losses ([USDA Risk Management Agency, 2021, 2023](#)).

The economic structure of LRP is closely related to put-option protection. A put option protects the holder against downside price movements while preserving upside price exposure. LRP provides similar price protection through a subsidized insurance contract. This link is consistent with the broader literature on hedging and commodity options ([Working, 1953](#); [Ederington, 1979](#); [Anderson and Danthine, 1983](#); [Vercammen, 1995](#)). It also motivates treating LRP as a market-linked insurance product.

LRP adoption has historically been limited relative to the size of the eligible cattle sector. Prior work identifies premium cost, producer familiarity, perceived usefulness, risk attitudes, product complexity, and other participation frictions as relevant factors ([Boyer and Griffith, 2023b](#)). Lower producer-paid premiums may therefore leave important nonprice frictions unchanged.

2.2 Premium Subsidies, Participation, and Coverage Choice

The producer-paid premium for LRP is reduced by a federal subsidy. USDA increased LRP support in 2019 and introduced additional program improvements in 2020, including more generous subsidy rates for higher coverage levels ([USDA Risk Management Agency, 2019, 2020](#)). These reforms reduced the cost of coverage and changed the relative price of choosing richer contracts. Prior work on LRP subsidy changes shows that the reforms were large enough to affect producer incentives ([Boyer and Griffith, 2023b](#)).

A key feature of LRP is that producers can respond to subsidies along several margins. They may enter the program, insure more head, increase insured liability, or select a higher coverage level. These margins have different policy meanings. Increases in insured head, policy counts, or liability indicate broader use of the program. Increases in coverage level or the share of insured head at high coverage levels indicate deeper protection among producers who already participate. Separating these margins is therefore central to evaluating whether subsidies expand access or mainly increase the generosity of coverage.

This distinction follows a broader literature on agricultural insurance subsidies. Crop insurance studies show that subsidies can affect participation, coverage choices, acreage decisions, and program costs ([Goodwin, 1993](#); [Goodwin and Smith, 1995](#); [Glauber, 2004](#); [Smith and Glauber, 2012](#); [Yu et al., 2018](#)). In crop insurance, acreage and coverage decisions are often the relevant margins. In livestock price insurance, the analogous margins are insured head, liability, policy counts, and coverage level. A subsidy that increases high-coverage LRP contracts may improve protection for active users, but it does not necessarily imply broader participation in the program.

2.3 Market-Linked Price Protection and CME Options

Although LRP is a public insurance product, its economic structure is linked to private livestock derivatives markets. Feeder cattle and live cattle futures provide market benchmarks for livestock price risk, and CME put options provide market prices for downside protection. Futures and options have long been studied as central instruments for managing commodity price risk ([Working, 1953](#); [Ederington, 1979](#); [Anderson and Danthine, 1983](#); [Moschini and Hennessy, 2001](#); [Vercammen, 1995](#)). Because LRP provides price protection against adverse livestock price movements, CME option prices are informative about the market cost of similar downside protection.

This institutional connection matters for the empirical design. Matched CME put option prices summarize the market cost of protection against price declines for the relevant cattle market, contract horizon, and coverage level. When put prices are high, the market value of downside protection is high. Since LRP provides a related form of protection, matched put prices are informative about the economic value of the insurance protection embedded in LRP contracts. At the same time, local county-level LRP demand shocks are unlikely to determine national CME option prices. This motivates using matched CME put prices as a source of variation in subsidy intensity.

The link between subsidized LRP and private derivatives also matters for policy interpretation. If subsidy increases mainly induce high-coverage purchases among producers already familiar with market-linked risk management, the policy effect differs from broad access expansion. Recent work on LRP subsidy capture raises a related concern. [Feuz \(2025\)](#) documents purchase patterns consistent with subsidy capture in livestock insurance, and related theoretical work argues that subsidized livestock price insurance can create incentives to extract subsidy rents through offsetting derivatives positions ([Adjemian and Ramsey, 2025](#)). This does not imply that all high-coverage demand reflects subsidy capture, but it reinforces the need to distinguish between broader participation and deeper coverage among existing users.

2.4 External Shocks and State-Contingent Demand

LRP covers livestock price declines. Drought losses and pandemic disruptions are outside its indemnity trigger, although these shocks can affect whether and how much insurance producers buy. Covid-era processing constraints and supply-chain disruptions changed livestock prices, meat-price spreads, and marketing conditions ([Lusk et al., 2021](#)). Drought affects pasture, hay availability, feed costs, herd management, and marketing timing. These channels connect the analysis to work on weather-index insurance and state-contingent climate risk ([Barnett and Mahul, 2007](#); [Mahul, 2001](#); [Chambers and Quiggin, 2000](#)).

The direction of the drought effect on LRP demand is theoretically ambiguous. State-

contingent models of production and insurance emphasize that producers' choices depend on the distribution of states they face and the ability of insurance to transfer resources across those states (Chambers and Quiggin, 2000; Mahul, 2001). Under this logic, drought could increase the value of price protection because producers face a more difficult risk environment. At the same time, severe drought can tighten cash flow, force earlier sales, reduce the number of head available to insure, or shift attention toward immediate production problems. In that case, drought can suppress insurance purchases even when risk protection is more valuable. This mechanism is consistent with work emphasizing downside risk and adverse shocks in agricultural decision making (Chavas, 2019).

We measure drought exposure using the U.S. Drought Monitor, which provides weekly county-level classifications from abnormally dry conditions to exceptional drought (Svoboda et al., 2002). Drought and Covid-era disruptions enter the analysis as external state-contingent factors. They may confound post-reform comparisons because the policy period overlaps with major market and production disruptions. Drought also provides a direct measure of production stress. Subsidies can lower the price of insurance and encourage high-tier coverage, while severe production stress may reduce participation and insured exposure.

3 Data

3.1 LRP endorsement data

The insurance data come from the United States Department of Agriculture Risk Management Agency administrative records for Livestock Risk Protection insurance. We read annual files for reinsurance years 2005 through 2026 and assign time using each endorsement's policy effective date. The resulting full-LRP sample begins in January 2005 and currently ends on July 24, 2025. Each record identifies a specific endorsement and reports its effective date, length, commodity, cattle type, location, coverage price, coverage level, insured head, premium, subsidy, liability, and indemnity. We retain feeder and fed cattle because these products are linked to active CME livestock derivatives markets.

We construct an independent full-LRP endorsement file and a county-quarter panel. County identifiers use the complete state and county FIPS codes reported by RMA, and calendar quarters use policy effective dates. County-quarter totals include insured head, endorsements, liability, premium, subsidy, and indemnities. Contract characteristics are weighted by insured head. Coverage-tier distributions are calculated separately at the endorsement level, so each insured head is assigned using its own endorsement's coverage level.

The main subsidy variable is subsidy intensity. It is defined as the subsidy amount per

dollar of insured liability,

$$\Gamma_{ict} = \frac{\text{Subsidy Amount}_{ict}}{\text{Liability}_{ict}}.$$

This measure captures the value of government support relative to the insured value of the contract. It differs from the statutory premium subsidy rate because it incorporates both premium rates and liability. We use this measure because it corresponds to the economic incentive created by the subsidy and follows the crop insurance literature on subsidy intensity (Yu et al., 2018). We also compute the subsidy share of total premium as a descriptive and robustness measure.

The main outcomes are designed to distinguish participation from contract generosity. The extensive margin outcomes are insured head, number of policies, and insured liability. These outcomes measure whether higher subsidies expand the scale of LRP use. The intensive margin outcomes are average coverage level and the share of insured head covered at coverage levels of at least 97 percent. These outcomes measure whether higher subsidies induce producers to select richer protection. This distinction is important because prior work shows that LRP adoption is shaped by risk preferences, familiarity, perceived usefulness, and other nonpremium frictions (Boyer et al., 2024). A subsidy may therefore deepen coverage without bringing many new producers into the program.

The LRP program changed materially during the sample period. USDA increased LRP premium subsidies in 2019 and introduced a more generous tiered schedule in 2020 (USDA Risk Management Agency, 2019, 2020; Boyer and Griffith, 2023a). We use these changes in descriptive period comparisons and event-study diagnostics. The main instrumental-variable design uses matched CME put premiums.

3.2 CME futures and put option data

The private market data come from CME livestock futures and options for live cattle and feeder cattle. Futures prices are observed at the contract month level and provide the market benchmark used to normalize option prices. Options data are transaction level CME livestock option trades. Each trade reports the trading date, option type, contract month, strike, transaction price, and volume. Because LRP provides downside price protection, the main analysis uses CME put options.

The role of CME option data in this paper is different from a direct price wedge analysis. We do not use the matched put premium as the dependent variable. Instead, we use the matched put price as an instrument for LRP subsidy intensity. The economic logic is that put option prices measure the market cost of downside price protection. When the market price of downside protection is higher, the actuarial value of LRP protection and

the subsidy per dollar of liability also change. This link follows from the close relationship between LRP, futures markets, and option based hedging ([Working, 1953](#); [Ederington, 1979](#); [Anderson and Danthine, 1983](#); [Vercammen, 1995](#)).

For each option trade, we construct a normalized put price,

$$Z_{mjt} = \log \left(\frac{P_{mjt}^{put}}{F_{mjt}} \right),$$

where P_{mjt}^{put} is the CME put transaction price and F_{mjt} is the corresponding futures price for cattle market j , contract month m , and trading date t . Normalizing by the futures price places option prices on a comparable scale across contracts and market conditions. The resulting variable measures the relative market cost of downside protection.

3.3 Drought, Covid, and county exposure data

We supplement the LRP endorsement data with county-level drought exposure from the U.S. Drought Monitor. The U.S. Drought Monitor provides weekly county-level drought classifications ranging from D0, abnormally dry, to D4, exceptional drought ([Svoboda et al., 2002](#)). The raw drought files are organized at the county-week level and report the share of county area in each drought category. We aggregate these weekly measures to the county-quarter level to match the LRP demand panel. For each county-quarter, we compute average exposure to D1-or-worse, D2-or-worse, D3-or-worse, and D4 drought conditions. The main drought analysis focuses on D4 exposure, scaled so that one unit corresponds to a 10 percentage point increase in the county area under exceptional drought.

Drought does not directly trigger LRP payments. We use it as an external measure of production stress. Pasture conditions, hay availability, feed costs, herd management, marketing timing, and liquidity can transmit drought into LRP demand. Weather-related risk has motivated a broader literature on agricultural insurance design and state-contingent risk management ([Barnett and Mahul, 2007](#); [Mahul, 2001](#); [Chambers and Quiggin, 2000](#)). Drought measures enter the matched-put robustness models as controls and sample restrictions. They are also the explanatory variables in a separate balanced county-quarter analysis.

We also account for Covid-era cattle market disruptions. Covid-period disruptions are relevant for cattle markets because processing capacity constraints and supply-chain disruptions created large short-run changes in livestock prices, meat price spreads, and marketing conditions ([Lusk et al., 2021](#)). In the robustness analysis, we use indicators for 2020 and 2021 and estimate specifications that exclude these years. These checks are not intended to identify a separate Covid effect. They assess whether the main subsidy estimates are sensitive to the overlap between the post-reform period and broader market

disruptions.

For heterogeneity analysis, we merge county-level cattle and hay measures from the USDA National Agricultural Statistics Service Quick Stats database ([USDA NASS, 2021](#)). We use Census of Agriculture county data for 2007, 2012, 2017, and 2022 to construct measures of cattle inventory, hay acreage, and hay acreage per 100 cattle. These variables are used to classify counties by livestock exposure and forage dependence. The NASS variables are not part of the main subsidy instrument. They are used only to examine whether drought effects are stronger in counties where cattle production and forage conditions are more economically relevant.

3.4 Matching procedure and estimation sample

We use the corrected endorsement-to-option matches constructed from cattle market, policy effective date, option maturity, and log moneyness. Moneyness is the option strike divided by the corresponding futures price. The main rule retains valid put matches with a maturity gap no larger than 30 days and an absolute log-moneyness gap no larger than 0.20:

$$|\Delta \text{Maturity}_e| \leq 30 \quad \text{and} \quad |\Delta \log \text{Moneyness}_e| \leq 0.20.$$

The strict robustness rule reduces these limits to 14 days and 0.10. The maturity restriction accounts for the horizon of LRP protection. The log-moneyness restriction compares the relative location of the option strike and the LRP price protection. Recalculated Black–76 implied volatility must be valid for an endorsement to enter the matched sample.

For each county, year-quarter, and cattle type, we calculate the insured-head-weighted mean of the normalized matched put premium. We also aggregate futures prices, moneyness, time to maturity, option volume, and matching gaps. The main instrument is the log of this mean normalized put premium. The alternative-instrument exercise uses the log matched Black–76 implied volatility.

The FEIV sample covers 2005 through 2024 and requires positive insured head, liability, and premium; nonmissing subsidy intensity and controls; and a valid matched put benchmark. The main rule retains 90.2 percent of eligible feeder endorsements and 97.9 percent of eligible fed endorsements. The strict rule retains 67.1 percent and 46.6 percent, respectively. Maturity is the more restrictive matching dimension in these data. Feeder cattle provide the larger estimation sample and form the primary analysis. Fed cattle estimates provide complementary evidence.

The matched CME put sample is the main sample for the FEIV subsidy estimates. Because this sample requires positive LRP activity and an observable same-day option-market

benchmark, it is not designed to measure county-quarters with no LRP purchases. For drought-specific analyses, we therefore construct a separate balanced county-quarter panel from the full LRP endorsement records. This balanced panel retains county quarters without LRP purchases as zero-demand observations for counties that appear in the LRP data. It allows us to study participation and insured scale directly. The balanced drought panel is used only for the external-shock and drought analyses; the main subsidy estimates remain based on the matched CME put sample.

4 Empirical Method

This section describes how we estimate the effect of LRP premium subsidies on insurance demand and how we evaluate whether external shocks, especially drought and Covid-era disruptions, affect the demand environment. The main design estimates the effect of subsidy intensity using matched CME put option prices as an instrument. We then conduct two sets of external-shock analyses. First, we reestimate the main FEIV specifications with drought controls and Covid-period exclusions in the matched CME put sample. Second, we construct a separate balanced county-quarter panel that retains inactive county-quarters and use it to estimate the direct association between drought exposure and LRP participation, insured scale, and coverage choice.

4.1 Demand margins and subsidy intensity

The empirical analysis separates participation and scale from contract generosity. Let i index counties, t index year-quarters, and c index cattle type. The extensive-margin outcomes are insured head, number of policies, and insured liability. These outcomes capture whether subsidies expand the scale of LRP use. The intensive-margin outcomes are average coverage level and the share of insured head covered at coverage levels of at least 97 percent. These outcomes capture whether subsidies induce producers to choose richer protection conditional on using LRP.

The main treatment variable is subsidy intensity, defined as the subsidy amount per dollar of insured liability,

$$\Gamma_{ict} = \frac{\text{Subsidy Amount}_{ict}}{\text{Liability}_{ict}}.$$

This measure captures public support relative to insured value. It differs from the statutory premium subsidy rate because it incorporates both premium rates and insured liability. This construction follows the crop insurance literature, where subsidy per unit of insured value is used to measure the economic incentive created by premium support (Yu et al., 2018).

4.2 Main FEIV design using matched CME put prices

Observed subsidy intensity may be endogenous. Producers facing greater perceived price risk may choose different coverage prices, endorsement lengths, or coverage levels, all of which affect the subsidy amount. Counties with stronger demand for risk protection may also respond differently to market conditions. To address this concern, we use variation in matched CME put option prices as an instrument for subsidy intensity.

For each endorsement, we use the corrected put-option match defined by cattle type, policy effective date, maturity, and relative log moneyness. The main limits are 30 maturity days and 0.20 absolute log-moneyness. We normalize the put premium by the corresponding futures price and aggregate it to the county-quarter-cattle-type level:

$$Z_{ict} = \log \left(\frac{P_{ict}^{put}}{F_{ict}} \right),$$

where P_{ict}^{put} is the matched CME put option price and F_{ict} is the corresponding futures price. The normalization makes option prices comparable across contracts and market conditions. The instrument measures the market cost of downside price protection.

The baseline first stage is

$$\log \Gamma_{ict} = \pi Z_{ict} + X'_{ict} \rho + \alpha_i + \lambda_t + u_{ict}.$$

The second stage is

$$Y_{ict} = \beta \widehat{\log \Gamma_{ict}} + X'_{ict} \delta + \alpha_i + \lambda_t + \varepsilon_{ict}.$$

The vector X_{ict} includes log coverage price, average endorsement length, matched futures price, moneyness, time to maturity, option volume, and the two matching-gap measures. County fixed effects α_i absorb persistent differences in cattle production, familiarity with LRP, and local risk-management practices. Year-quarter fixed effects λ_t absorb national seasonality, common policy changes, livestock-market conditions, and macroeconomic shocks. The baseline standard errors are clustered by county. State clustering is reported as a robustness check.

When Y_{ict} is logged insured head, endorsements, or liability, β is an elasticity with respect to subsidy intensity. For log average coverage, it is also an elasticity. For the high-coverage share, it measures the percentage-point-equivalent change in the 0–1 share associated with a one-log-point change in subsidy intensity.

The identifying assumption is that matched CME put premiums affect county-level LRP

demand through subsidy intensity after conditioning on fixed effects and observed contract and market characteristics. Local LRP demand is unlikely to move national CME option prices. A remaining concern is that contemporaneous endorsement choices affect which option is selected. Controls for coverage price, endorsement length, maturity, moneyness, futures prices, option volume, and matching gaps address observable composition. The strict matching rule, alternative implied-volatility instrument, cattle-type splits, sample restrictions, and external-shock controls provide additional diagnostics. The instrument is a matched national option-market price. It is not a shift-share instrument and does not use a post-reform indicator interacted with county exposure.

4.3 External-shock robustness in the matched CME put sample

The post-reform period overlaps with broader shocks to cattle markets and production conditions, including Covid-era disruptions and severe drought episodes. These shocks are not the source of identification in the main design, but they may affect the demand environment. We therefore reestimate the main FEIV specifications after adding drought controls and after excluding periods affected by Covid and severe drought.

The augmented first stage is

$$\log \Gamma_{ict} = \pi Z_{ict} + X'_{ict}\rho + W'_{it}\eta + \alpha_i + \lambda_t + u_{ict},$$

and the augmented second stage is

$$Y_{ict} = \beta \widehat{\log \Gamma}_{ict} + X'_{ict}\delta + W'_{it}\phi + \alpha_i + \lambda_t + \varepsilon_{ict}.$$

The vector W_{it} contains county-quarter drought exposure measures from the U.S. Drought Monitor. In the robustness specifications, drought controls include severe and extreme drought exposure measured at the county-quarter level. We also estimate specifications with one-quarter lagged drought controls. Covid-period robustness is implemented by excluding 2020 and 2021. Additional checks exclude severe drought quarters, and exclude both Covid-period years and severe drought quarters.

These specifications answer a narrow question: whether the main subsidy estimates are sensitive to contemporaneous drought exposure, lagged drought exposure, or Covid-era sample restrictions. They do not treat drought or Covid as instruments for subsidy intensity.

4.4 Balanced county-quarter drought panel

The matched CME put sample is appropriate for the main FEIV design because it requires positive LRP activity and an observable same-day option-market benchmark. However, that sample is not designed to study inactive county-quarters. To examine whether drought affects participation and insured scale directly, we construct a separate balanced county-quarter panel from the full LRP endorsement records.

The balanced drought panel includes county-by-cattle-type cells with at least one LRP purchase during the sample period and retains county-quarters without purchases as zero-demand observations. This construction allows us to study whether drought changes the probability that a county-quarter has any LRP activity. The drought panel is constructed for 2005 through 2024, matching the period covered by the county-level U.S. Drought Monitor data used in the analysis.

For the drought analysis, the main outcomes are:

$$\mathbf{1}\{\text{Active}_{ict} > 0\}, \quad \log(1 + \text{Policies}_{ict}), \quad \log(1 + \text{Head}_{ict}), \quad \log(1 + \text{Liability}_{ict}).$$

The $\log(1 + y)$ transformation allows us to retain zero-demand county-quarters. We also estimate conditional coverage outcomes among active county-quarters, including log average coverage level and the high-coverage share. These conditional outcomes distinguish drought effects on participation and scale from drought effects on coverage choice among users.

Weekly U.S. Drought Monitor measures are aggregated to the county-quarter level. Let D_{it}^k denote the average share of county area in drought category k or worse during quarter t . We examine D1-or-worse, D2-or-worse, D3-or-worse, and D4 exposure. The main drought variable is D4 exposure, scaled so that one unit equals a 10 percentage point increase in county area under exceptional drought.

For each drought threshold, we estimate

$$Y_{ict} = \theta D_{it}^k + \alpha_i + \lambda_t + \nu_{ict},$$

where α_i are county fixed effects and λ_t are year-quarter fixed effects. Standard errors are clustered at the county level. These specifications identify drought associations from within-county changes in drought exposure over time, net of national quarter shocks.

We also estimate models with county-specific linear trends to separate D4 exposure from long-run regional movement.

$$Y_{ict} = \theta D_{it}^k + \alpha_i + \lambda_t + \tau_i t + \nu_{ict}.$$

The coefficient θ in these models compares demand outcomes within the same county over time, after allowing each county to follow its own linear trend.

4.5 Drought margins, heterogeneity, and option-market comparison

The drought analysis proceeds in three steps. First, we estimate threshold models for D1-or-worse, D2-or-worse, D3-or-worse, and D4 exposure. This tests whether LRP demand responds gradually to drought severity or only under extreme conditions. Second, we decompose the D4 effect across participation, policy counts, insured head, liability, and conditional coverage choices. This distinguishes drought effects on the breadth and scale of LRP use from drought effects on coverage generosity among active users.

Third, we examine heterogeneity using county-level cattle and hay measures from USDA NASS Quick Stats. We classify counties by cattle inventory, hay acreage, hay acreage per 100 cattle, and related cattle-exposure measures. We then interact D4 exposure with indicators for high cattle exposure or high forage dependence. These specifications test whether drought effects are stronger in counties where cattle production and forage conditions are more economically relevant.

Finally, we compare LRP demand with matched CME option activity in the smaller matched-option sample. This comparison is used only as a mechanism check. The goal is to assess whether severe drought appears to shift demand from LRP into CME option activity. Because matched option outcomes are observed only for endorsements that can be linked to option-market information, these estimates are not interpreted as full CME market demand.

5 Results

5.1 Subsidy intensity, participation, and coverage choice

Table 1 describes the corrected analysis-ready panels before fixed-effect singleton removal. The main rule contains 12,608 feeder county-quarters and 2,199 fed county-quarters. The strict rule contains 11,331 and 1,749 observations. The corresponding FEIV models retain 12,578 and 2,164 observations under the main rule and 11,294 and 1,692 under the strict rule. Feeder cattle provide substantially more cross-county and time variation, so the main discussion focuses on that market.

Table 2 reports the feeder-cattle FEIV estimates. Log normalized matched put premium

instruments for log subsidy intensity. The first-stage coefficient is 0.664 with a county-clustered standard error of 0.020. The excluded-instrument Wald statistic is 1089.3. This statistic equals the square of the cluster-robust first-stage t statistic because the model has one excluded instrument.

The strongest extensive-margin response is the number of endorsements. Its elasticity is 0.505 with a standard error of 0.060. A 10 percent increase in subsidy intensity therefore predicts approximately 5 percent more endorsements. The insured-head coefficient is 0.197, and the liability coefficient is 0.189. Both are positive and marginally significant in the main rule.

Coverage composition also responds. The coefficient on the share of insured head at coverage levels of at least 97 percent is 0.208 with a standard error of 0.027. A 10 percent increase in subsidy intensity predicts an increase of approximately 1.98 percentage points in this share. The coefficient on log average coverage is 0.0029 and is significant only at the 10 percent level. Average coverage therefore provides weaker evidence than movement into the highest tier.

Table 3 applies the strict 14-day and 0.10 log-moneyness limits. The endorsement-count elasticity remains close to the main estimate at 0.519. The high-tier share coefficient remains positive and significant at 0.172. Under the strict rule, insured head and liability rise by 0.301 and 0.283, respectively, and both reach the 5 percent significance level. Average coverage becomes small and insignificant. The two matching rules agree on transaction growth and movement into the highest coverage tier. Evidence on total insured scale is positive but sensitive to the matching and time samples.

Figure 1 describes dynamics around the 2020Q3 tiered subsidy reform. Joint tests reject parallel pre-trends for endorsement counts, the high-tier share, and subsidy intensity. The corresponding p -values are 0.0075, 0.0013, and 0.0052. Insured head and liability do not show comparable rejection. We use this exercise as a diagnostic and do not interpret it as an independent policy experiment.

5.2 Transactions and coverage composition

Table 4 and Figure 2 summarize coverage using the independent full-LRP endorsement file. Each endorsement is placed in one of three bins using its own coverage level. The share of feeder insured head at coverage levels of at least 97 percent was 54.6 percent during 2005–2018. It rose to 86.5 percent during 2019–2021 and 90.9 percent in the available 2022–2025 records. The below-95-percent share fell from 32.8 percent to 5.5 percent. These distributions document a substantial composition change, although the event-study pre-trends prevent a simple causal before-and-after interpretation.

Subsidy intensity is associated with more endorsements and greater use of the highest

tier. These margins can move together. More transactions do not necessarily imply entry by previously uninsured producers because the administrative data do not contain stable producer identifiers. The balanced county analysis offers a diagnostic for local participation, but its reform interactions also exhibit pre-existing trends. We therefore describe the policy-count result as transaction growth and reserve claims about producer entry for future producer-level data.

Figure 3 and Appendix Tables 30–31 document spatial concentration using the full-LRP panel. South Dakota, Nebraska, Texas, Oklahoma, and Kansas account for about 60.6 percent of feeder insured head. This pattern is consistent with the geography of cattle production and established LRP distribution networks. Leave-one-state-out estimates show that no single large state explains the high-tier FEIV result. Its coefficient remains between 0.190 and 0.218 when each of the twelve largest states is removed.

5.3 Drought, Covid, and state-contingent demand

Appendix Table 10 reestimates the corrected FEIV models with current drought, lagged drought, Covid-period exclusions, and drought-based sample restrictions. Endorsement counts and the high-tier share remain positive across these checks. Insured head and liability vary more with the sample. The short-control estimates are much larger than the fully controlled estimates, especially for coverage outcomes. We treat this difference as evidence that contract and option-matching controls are substantively important.

Second, we study drought directly using a separate balanced county-quarter panel constructed from the full LRP endorsement records and merged with county-level U.S. Drought Monitor exposure. This panel retains inactive county-quarters as zero-demand observations, which makes it better suited for studying participation and insured scale. The drought specifications include county fixed effects and year-quarter fixed effects, with standard errors clustered by county. The main drought variable is the county-area share under exceptional drought, D4, scaled so that one unit represents a 10 percentage point increase in exposure.

The drought estimates indicate a nonlinear participation constraint for feeder cattle. Mild and moderate drought exposure has little systematic relationship with LRP use. The clearest response appears under exceptional drought. In Appendix Table 11, a 10 percentage point increase in D4 exposure is associated with a 0.303 percentage point lower probability that a county-quarter has any feeder-cattle LRP activity, a 2.0 percent reduction in insured head, a 0.8 percent reduction in the number of policies, and a 4.0 percent reduction in liability. These estimates are very similar after adding county-specific linear trends: Appendix Table 12 reports a D4 coefficient of -0.0202 in the baseline log insured-head specification and -0.0205 with county trends. In economic terms, full-county

D4 exposure is associated with about 18 percent lower insured head and 33 percent lower liability, as shown in Appendix Table 14.

The margin decomposition connects drought to the subsidy results. Appendix Table 13 shows lower activity and insured scale during D4 exposure. Among active county-quarters, average coverage and the high-tier share change little. The same pattern appears after adding county-specific linear trends. Extreme production stress is therefore associated with contraction in insurable activity. It does not produce a detectable change in coverage composition among continuing users.

Heterogeneity results provide some evidence that the D4 response is stronger in cattle-intensive counties, though the pattern is not uniform across all exposure definitions. Appendix Table 15 shows that the additional D4 effect in top-quartile cattle-inventory counties is negative and statistically significant in the county fixed-effects specification, with an interaction coefficient of about -0.027 . The corresponding county-trend estimate remains negative but is only marginally significant. Other definitions, including top-decile cattle inventory and joint high-cattle/high-LRP-use classifications, are less precise. We therefore treat the heterogeneity evidence as suggestive rather than definitive.

Finally, Appendix Table 23 compares LRP activity with matched CME option activity during severe drought in the smaller matched sample. The results do not show substitution from LRP into exchange-traded options. Under D3-plus drought exposure, feeder-cattle LRP insured head, liability, and policy counts decline, and matched option volume also tends to decline. This pattern is more consistent with a general tightening of risk-management activity under severe drought than with producers replacing LRP with private option positions.

Taken together, the external-shock results separate two demand margins. Subsidy intensity increases endorsement counts and the high-tier share. Exceptional drought is associated with lower participation and insured exposure. The D4 onset estimates contain significant pre-onset movement, so they do not identify an abrupt causal effect. The county fixed-effects and county-trend results nevertheless provide consistent evidence of contraction during exceptional drought.

6 Conclusion

This paper studies how LRP subsidy intensity affects transactions, insured scale, and coverage composition. Corrected maturity and moneyness matches link RMA endorsements to CME put premiums. The main FEIV sample covers 2005–2024, while the independent full-LRP panel uses effective dates observed from January 2005 through July 2025.

Two feeder-cattle results are stable. Higher subsidy intensity increases endorsement counts

and shifts insured head into coverage levels of at least 97 percent. A 10 percent increase in subsidy intensity predicts about 5 percent more endorsements and a 1.6 to 2.0 percentage point increase in the high-tier head share. Insured head and liability also have positive coefficients, although significance depends on the matching rule and time sample. Average coverage changes little and is not a robust outcome.

These findings indicate growth in LRP transactions and a change in coverage composition. They do not establish how many previously uninsured producers entered because the public endorsement data lack stable producer identifiers. The full-LRP records show that use remains geographically concentrated. Five major cattle states account for about 61 percent of feeder insured head. The high-tier FEIV coefficient remains stable when these states are excluded one at a time.

Exceptional drought is associated with lower feeder-cattle activity, insured head, endorsement counts, and liability. Coverage choices among active county-quarters remain largely unchanged. The onset analysis shows that insured head begins falling before first D4 classification. We therefore interpret the drought estimates as within-county conditional associations. They are consistent with herd adjustment, earlier marketing, liquidity pressure, and contraction in the cattle exposure available for insurance.

The results also speak to recent concerns about the interaction between public livestock insurance and private derivatives markets. LRP is institutionally linked to CME livestock markets, and recent studies raise concerns that subsidized livestock price insurance can create opportunities for subsidy capture or subsidy harvesting ([Feuz, 2025](#); [Adjemian and Ramsey, 2025](#)). Our analysis does not test individual trading strategies or identify subsidy capture directly. It does show that subsidy-induced demand responses are concentrated in high-coverage contracts. That pattern is relevant for policy design because high-coverage contracts are where richer public support, stronger downside protection, and potential interactions with private derivatives markets become most salient.

Several limitations remain. The instrumental variable design uses matched CME put option prices as market-based variation in the cost of downside protection. This approach is motivated by the institutional link between LRP and livestock derivatives markets, but it identifies effects for observations with observable matched option benchmarks. The estimates should therefore be interpreted as applying to the matched LRP market rather than to all potential cattle producers. In addition, the analysis is conducted at the county-quarter level and cannot observe individual producer balance sheets, risk preferences, agent relationships, or private derivatives positions. These limitations prevent us from making direct welfare claims about individual producers or from measuring subsidy capture behavior at the trader level.

The policy implications depend on the intended margin. Subsidies appear effective at

increasing transactions and use of the highest coverage tier. Evidence on total insured scale is less stable. Expanding access may also require information, agent capacity, liquidity support, and greater familiarity with market-linked insurance. Producer-level enrollment histories would allow future work to distinguish new entry from repeated endorsements and to examine the interaction between LRP and private hedging positions.

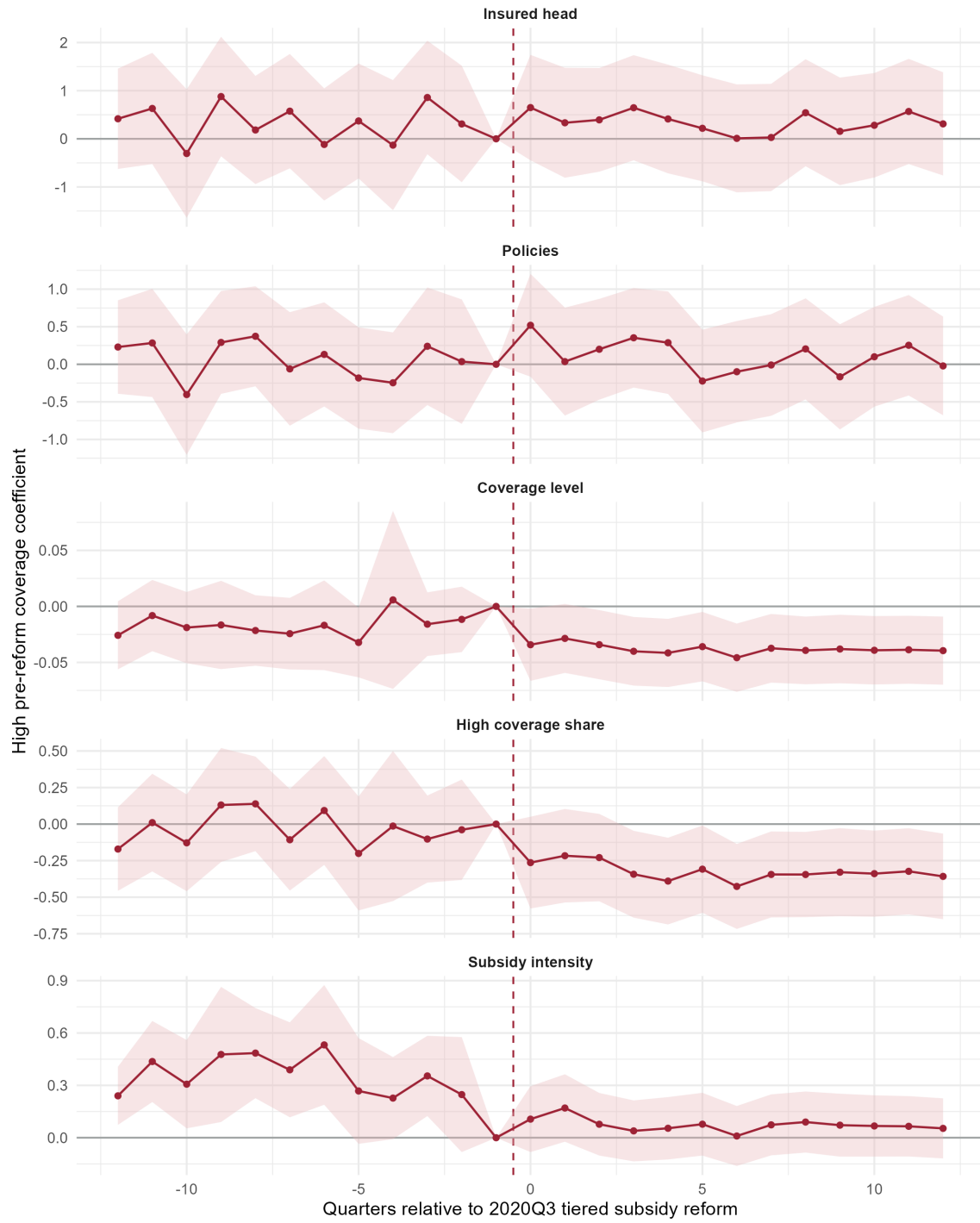


Figure 1: Event study evidence around the 2020 tiered subsidy reform

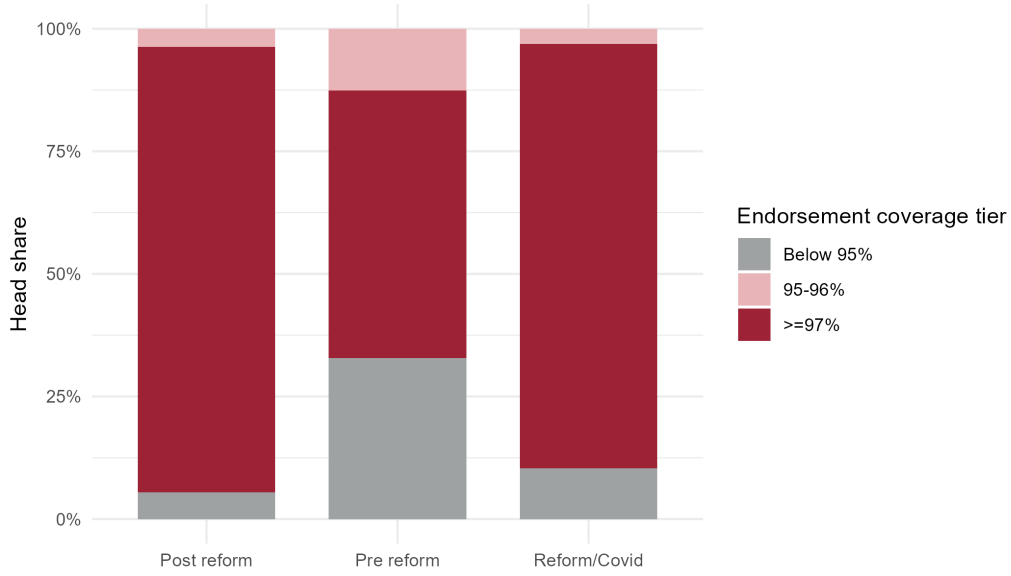


Figure 2: Coverage tier distribution before and after the subsidy reforms

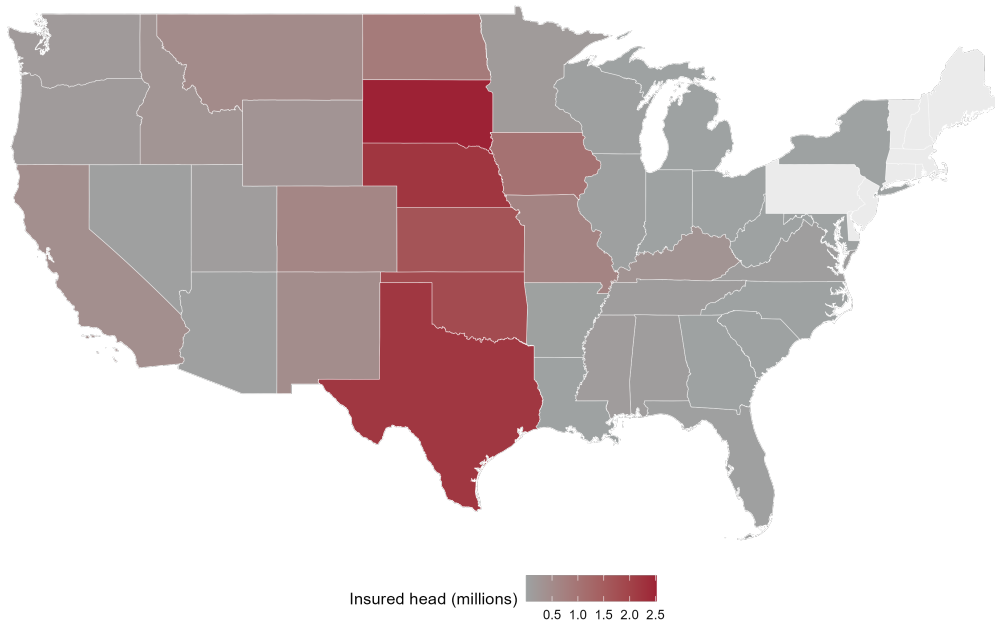


Figure 3: Spatial concentration of feeder cattle LRP insured head, January 2005–July 2025

Table 1: Corrected matched-sample description

Matching	Cattle	Observations	Counties	States	Mean head	Mean policies	Mean liability (\$m)	Mean share $\geq 97\%$
Main 30d/0.20	Feeder	12,608	983	39	918.8	7.0	1.50	0.746
Main 30d/0.20	Fed	2,199	325	25	1,118.6	5.3	2.54	0.770
Strict 14d/0.10	Feeder	11,331	981	39	998.3	7.6	1.63	0.759
Strict 14d/0.10	Fed	1,749	306	23	1,332.9	6.3	3.04	0.778

Notes: The main rule requires a maturity gap no larger than 30 days and an absolute log-moneyness gap no larger than 0.20. The strict rule uses 14 days and 0.10. Statistics describe the finite-valued analysis-ready panels before fixed-effect singleton removal. Regression tables report the resulting model-specific observations.

Table 2: Main FEIV estimates for feeder cattle

Outcome	Estimate	Observations	Excluded-instrument F
ln insured head	0.197* (0.108)	12,578	1089.3
ln endorsements	0.505*** (0.060)	12,578	1089.3
ln liability	0.189* (0.111)	12,578	1089.3
ln average coverage	0.003* (0.002)	12,578	1089.3
Share $\geq 97\%$	0.208*** (0.027)	12,578	1089.3

Notes: The endogenous variable is log subsidy intensity. The instrument is the log normalized matched CME put premium. Models include county and year-quarter fixed effects and the full set of contract and option-matching controls. Standard errors in parentheses are clustered by county. The reported first-stage statistic is the clustered Wald F for the excluded instrument.

Table 3: Robustness to corrected option-matching rules

Matching rule	Outcome	Estimate	Observations	Excluded-instrument F
Main 30d/0.20	ln insured head	0.197* (0.108)	12,578	1089.3
Main 30d/0.20	ln endorsements	0.505*** (0.060)	12,578	1089.3
Main 30d/0.20	ln liability	0.189* (0.111)	12,578	1089.3
Main 30d/0.20	ln average coverage	0.003* (0.002)	12,578	1089.3
Main 30d/0.20	Share $\geq 97\%$	0.208*** (0.027)	12,578	1089.3
Strict 14d/0.10	ln insured head	0.301** (0.128)	11,294	766.4
Strict 14d/0.10	ln endorsements	0.519*** (0.080)	11,294	766.4
Strict 14d/0.10	ln liability	0.283** (0.133)	11,294	766.4
Strict 14d/0.10	ln average coverage	0.001 (0.002)	11,294	766.4
Strict 14d/0.10	Share $\geq 97\%$	0.172*** (0.032)	11,294	766.4

Notes: Each row reports a separate feeder-cattle FEIV model. The main and strict rules change the maximum maturity and absolute log-moneyness gaps. Standard errors are clustered by county.

Table 4: Endorsement-level feeder-cattle coverage tiers

Period	Coverage tier	Insured head	Endorsements	Head share
2005–2018	Below 95%	462,277	4,878	32.8%
2005–2018	95–96%	177,126	2,182	12.6%
2005–2018	At least 97%	769,093	9,218	54.6%
2019–2021	Below 95%	150,293	966	10.4%
2019–2021	95–96%	45,240	449	3.1%
2019–2021	At least 97%	1,253,844	8,675	86.5%
2022–2025 available	Below 95%	795,898	7,272	5.5%
2022–2025 available	95–96%	532,890	5,284	3.7%
2022–2025 available	At least 97%	13,194,605	100,736	90.9%

Notes: Each endorsement is assigned to a tier using its own coverage level. Head shares are calculated from endorsement-level insured head. The final period contains policy effective dates observed through July 24, 2025. These are descriptive distributions.

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A Supplementary Results

This appendix reports supplementary analyses for the corrected matched-put design. It includes event-study diagnostics, robustness checks, drought estimates, heterogeneity, counterfactual calculations, and spatial evidence. These analyses interpret and test the main findings. They do not replace the baseline FEIV specification.

A.1 Event study diagnostics

Table 5 reports joint pre-trend tests around the 2020Q3 tiered subsidy reform. Insured head and liability show weaker evidence of differential pre-reform movement. Endorsement counts, the high-tier share, and subsidy intensity reject parallel pre-trends. We therefore use the event study as a diagnostic and rely on the matched-option FEIV design for the main estimates.

Table 5: Pre-reform joint tests

Outcome	F statistic	p-value	Observations
ln insured head	1.223	0.2711	8,800
ln endorsements	2.394	0.0075	8,800
ln liability	1.318	0.2136	8,800
ln average coverage	1.578	0.1046	8,800
Share $\geq 97\%$	2.891	0.0013	8,800
NA	2.501	0.0052	8,782

Notes: The event time is centered on 2020Q3, with quarter -1 omitted. The tests jointly assess the displayed pre-reform coefficients. County-clustered inference is reported.

A.2 Robustness checks

Table 6 varies clustering, controls, and sample restrictions. Endorsement counts and the high-tier share remain positive across the principal specifications. Insured head and liability are more sample dependent. The short-controls specification produces unusually large coverage coefficients, showing that contract and option-market controls are important.

A.3 Matching and first-stage diagnostics

Tables 7 and 8 report matching rates and first-stage estimates. Table 9 uses matched Black-76 implied volatility as the alternative instrument.

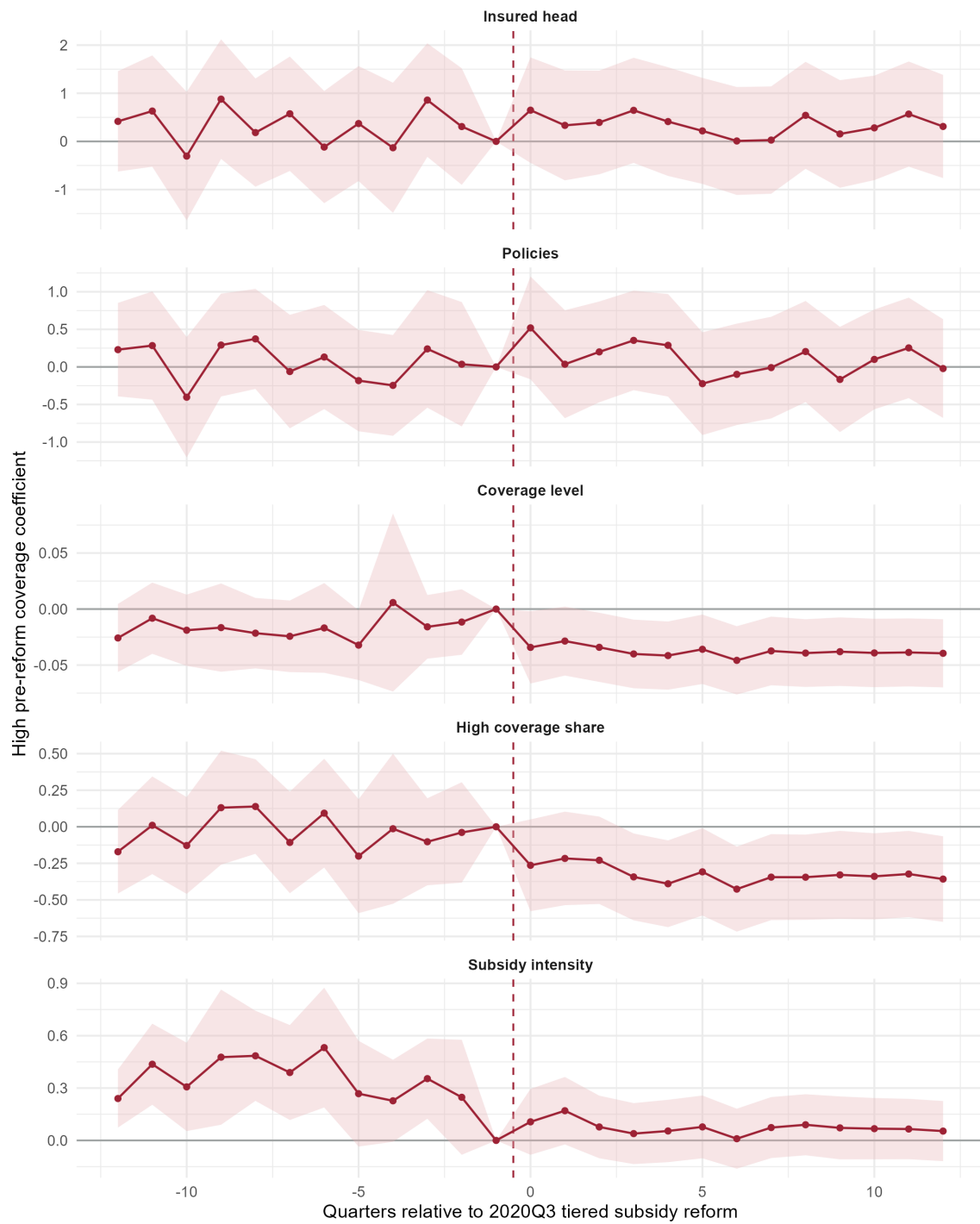


Figure 4: Event study estimates around the 2020 tiered subsidy reform

Table 6: Sample, clustering, and controls robustness

Specification	Outcome	Estimate	Observations	Excluded-instrument F
baseline	ln insured head	0.197* (0.108)	12,578	1089.3
baseline	ln endorsements	0.505*** (0.060)	12,578	1089.3
baseline	ln liability	0.189* (0.111)	12,578	1089.3
baseline	ln average coverage	0.003* (0.002)	12,578	1089.3
baseline	Share $\geq 97\%$	0.208*** (0.027)	12,578	1089.3
state cluster	ln insured head	0.197* (0.109)	12,578	3668.0
state cluster	ln endorsements	0.505*** (0.082)	12,578	3668.0
state cluster	ln liability	0.189* (0.104)	12,578	3668.0
state cluster	ln average coverage	0.003 (0.002)	12,578	3668.0
state cluster	Share $\geq 97\%$	0.208*** (0.021)	12,578	3668.0
short controls	ln insured head	0.254*** (0.070)	12,578	2303.0
short controls	ln endorsements	0.301*** (0.042)	12,578	2303.0
short controls	ln liability	0.313*** (0.073)	12,578	2303.0
short controls	ln average coverage	0.068*** (0.001)	12,578	2303.0
short controls	Share $\geq 97\%$	0.877*** (0.020)	12,578	2303.0
post 2015	ln insured head	0.096 (0.152)	9,596	1065.4
post 2015	ln endorsements	0.516*** (0.085)	9,596	1065.4
post 2015	ln liability	0.094 (0.158)	9,596	1065.4
post 2015	ln average coverage	0.003 (0.002)	9,596	1065.4
post 2015	Share $\geq 97\%$	0.169*** (0.035)	9,596	1065.4
exclude 2020 2021	ln insured head	0.244** (0.109)	10,822	957.4
exclude 2020 2021	ln endorsements	0.509*** (0.061)	10,822	957.4
exclude 2020 2021	ln liability	0.239** (0.113)	10,822	957.4
exclude 2020 2021	ln average coverage	0.003* (0.002)	10,822	957.4
exclude 2020 2021	Share $\geq 97\%$	0.220*** (0.029)	10,822	957.4

Notes: All models use the corrected main matching rule. The state-cluster specification clusters by state; the remaining specifications cluster by county. The short-controls estimates are reported as a sensitivity diagnostic.

Table 7: Corrected option-matching diagnostics

Rule	Cattle type	Eligible endorsements	Admissible matches	Match rate	Counties
Main 30d/0.20	feeder	89,424	80,677	90.2%	985
Main 30d/0.20	fed	11,739	11,490	97.9%	325
Strict 14d/0.10	feeder	89,424	59,978	67.1%	985
Strict 14d/0.10	fed	11,739	5,474	46.6%	325

Notes: The main rule permits a 30-day maturity gap and a 0.20 absolute log-moneyness gap. The strict rule permits 14 days and 0.10.

Table 8: Corrected first-stage estimates

Matching rule	Cattle type	Put-premium coefficient	Robust t	Excluded-instrument F	Observations
Main 30d/0.20	feeder	0.664 (0.020)	33.00	1089.3	12,578
Main 30d/0.20	fed	0.419 (0.031)	13.50	182.3	2,164
Strict 14d/0.10	feeder	0.587 (0.021)	27.68	766.4	11,294
Strict 14d/0.10	fed	0.414 (0.042)	9.96	99.3	1,692

Notes: The dependent variable is log subsidy intensity. Models include county and year-quarter fixed effects and full controls. Inference is clustered by county.

Table 9: Black-76 implied-volatility instrument robustness

Matching rule	Outcome	Estimate	Observations	Excluded-instrument F
main	ln insured head	-0.358* (0.196)	12,578	638.4
main	ln endorsements	0.296*** (0.112)	12,578	638.4
main	ln liability	-0.432** (0.203)	12,578	638.4
main	ln average coverage	0.003 (0.003)	12,578	638.4
main	Share $\geq 97\%$	0.138*** (0.044)	12,578	638.4
strict	ln insured head	-0.351 (0.257)	11,294	383.8
strict	ln endorsements	0.346** (0.158)	11,294	383.8
strict	ln liability	-0.428 (0.268)	11,294	383.8
strict	ln average coverage	-0.004 (0.003)	11,294	383.8
strict	Share $\geq 97\%$	0.057 (0.055)	11,294	383.8

Notes: These models replace normalized put premium with log matched Black-76 implied volatility. This is an alternative market-price construct.

A.4 Drought and Covid period robustness

Table 10 reports robustness checks for drought exposure and the Covid period in the corrected matched-put sample. The specifications assess whether broader cattle-market and production shocks account for the estimates.

Endorsement counts and the high-tier share remain positive after adding contemporaneous or lagged drought controls and under the Covid and drought sample exclusions. Insured head and liability remain more sensitive to the estimation sample. Average coverage does not provide a consistent main result.

A.5 Drought stress and state-contingent demand

The robustness checks above keep the matched-put sample fixed. A separate balanced county-quarter panel includes inactive observations and is suited to participation and insured-scale outcomes. We merge this panel with county-level U.S. Drought Monitor exposure. These estimates describe the demand environment under severe climate stress and do not identify subsidy elasticity.

Table 11 reports the basic drought-margin decomposition for feeder cattle. Mild and moderate drought exposure is not the main margin. The clearest and most stable negative response appears under exceptional drought, with D4 exposure reducing the probability of an active LRP county-quarter, the number of policies, insured head, and liability. Table 12 shows that the D4 result is not removed by adding county-specific linear trends.

Table 13 separates participation and scale from coverage choices among active feeder county-quarters. D4 exposure is associated with lower activity and insured scale. Average coverage and the high-tier share change little among continuing users.

Table 14 reports the economic magnitude of the D4 coefficients. Full-county exposure is

Table 10: Drought and Covid-period robustness in the corrected matched sample

Specification	Outcome	Estimate	Observations	Excluded-instrument F
baseline	ln insured head	0.197* (0.108)	12,578	1089.3
baseline	ln liability	0.189* (0.111)	12,578	1089.3
baseline	ln average coverage	0.003* (0.002)	12,578	1089.3
baseline	Share $\geq$ 97%	0.208*** (0.027)	12,578	1089.3
current drought	ln insured head	0.195* (0.108)	12,578	1091.7
current drought	ln liability	0.187* (0.111)	12,578	1091.7
current drought	ln average coverage	0.003* (0.002)	12,578	1091.7
current drought	Share $\geq$ 97%	0.208*** (0.027)	12,578	1091.7
lagged drought	ln insured head	0.213* (0.111)	11,570	976.5
lagged drought	ln liability	0.205* (0.115)	11,570	976.5
lagged drought	ln average coverage	0.003* (0.002)	11,570	976.5
lagged drought	Share $\geq$ 97%	0.211*** (0.029)	11,570	976.5
exclude covid	ln insured head	0.244** (0.109)	10,822	957.4
exclude covid	ln liability	0.239** (0.113)	10,822	957.4
exclude covid	ln average coverage	0.003* (0.002)	10,822	957.4
exclude covid	Share $\geq$ 97%	0.220*** (0.029)	10,822	957.4
exclude severe	ln insured head	0.182 (0.149)	7,588	716.3
exclude severe	ln liability	0.170 (0.154)	7,588	716.3
exclude severe	ln average coverage	0.002 (0.002)	7,588	716.3
exclude severe	Share $\geq$ 97%	0.204*** (0.036)	7,588	716.3
exclude covid severe	ln insured head	0.268* (0.158)	6,252	612.4
exclude covid severe	ln liability	0.267 (0.164)	6,252	612.4
exclude covid severe	ln average coverage	0.002 (0.002)	6,252	612.4
exclude covid severe	Share $\geq$ 97%	0.219*** (0.038)	6,252	612.4

Notes: All models use the corrected main matched-put instrument. Drought specifications add current or lagged severe and extreme drought exposure. Standard errors and first-stage Wald statistics are clustered by county.

Table 11: Feeder cattle demand margins under extreme drought

Outcome	D3 plus	D4	Observations
Active quarter	-0.001 (0.001)	-0.003*** (0.001)	87,120
ln insured head	-0.006 (0.004)	-0.020*** (0.007)	87,120
ln liability	-0.012 (0.009)	-0.040*** (0.014)	87,120
ln policies	-0.002* (0.001)	-0.008*** (0.002)	87,120

Each cell reports a separate feeder cattle county FE and year-quarter FE regression. Drought coefficients are measured for a 10 percentage point increase in county area under the indicated threshold. Standard errors are clustered by county.

Table 12: County-specific trend robustness for feeder cattle

Specification	Drought coefficient	Observations
Baseline D4	-0.020*** (0.007)	87,120
County trends D4	-0.020*** (0.006)	87,120
Baseline D3 plus	-0.006 (0.004)	87,120
County trends D3 plus	-0.010*** (0.004)	87,120

The dependent variable is log one plus insured head. Coefficients are measured for a 10 percentage point increase in county drought exposure. County trends are estimated using county-specific linear trends. Standard errors are clustered by county.

Table 13: Exceptional drought and LRP demand margins

Outcome	County FE	County trends	Observations
Active quarter	-0.003*** (0.001)	-0.003*** (0.001)	86,720
ln policies	-0.008*** (0.002)	-0.008*** (0.002)	86,720
ln insured head	-0.020*** (0.007)	-0.021*** (0.006)	86,720
ln liability	-0.040*** (0.014)	-0.041*** (0.013)	86,720
ln coverage level conditional active	-0.000 (0.000)	0.000 (0.000)	13,765
share high coverage conditional active	-0.003 (0.003)	-0.000 (0.003)	13,765

The table separates participation and scale outcomes from conditional coverage choice among active feeder cattle LRP county quarters. D4 coefficients are measured for a 10 percentage point increase in county area under exceptional drought. Standard errors are clustered by county.

associated with an active-quarter probability about 3 percentage points lower, insured head about 18 percent lower, and liability about 33 percent lower. These are conditional associations from county and year-quarter fixed-effects models.

Table 14: Economic magnitude of D4 drought exposure

Outcome	D4 exposure change	Effect	Baseline mean	Implied level change
Active quarter	10 pp D4 exposure	-0.30 [-0.51, -0.10]	0.16	-0.00
Active quarter	50 pp D4 exposure	-1.52 [-2.54, -0.50]	0.16	-0.02
Active quarter	100 pp D4 exposure	-3.03 [-5.07, -0.99]	0.16	-0.03
ln policies	10 pp D4 exposure	-0.79 [-1.16, -0.42]	1.16	-0.01
ln policies	50 pp D4 exposure	-3.89 [-5.66, -2.08]	1.16	-0.05
ln policies	100 pp D4 exposure	-7.62 [-10.99, -4.12]	1.16	-0.09
ln insured head	10 pp D4 exposure	-2.00 [-3.26, -0.72]	151.23	-3.02
ln insured head	50 pp D4 exposure	-9.60 [-15.26, -3.57]	151.23	-14.52
ln insured head	100 pp D4 exposure	-18.28 [-28.20, -7.00]	151.23	-27.65
ln liability	10 pp D4 exposure	-3.96 [-6.57, -1.29]		
ln liability	50 pp D4 exposure	-18.31 [-28.80, -6.28]		
ln liability	100 pp D4 exposure	-33.27 [-49.31, -12.16]		

Effects are based on the county fixed-effects decomposition estimates. For log outcomes, the table reports $100 \times [\exp(\beta \times \text{exposure units}) - 1]$. Because D4 exposure is scaled by 10 percentage points in the regressions, 100 pp D4 exposure corresponds to ten exposure units.

Tables 15, 16, and 17 provide additional diagnostics. The negative D4 response is stronger in top-quartile cattle-inventory counties, although other exposure definitions are less consistent. The decline is visible during one- and two-quarter D4 episodes. The onset event study shows significant movement before first D4 classification and serves only as timing evidence.

Finally, Table 23 compares LRP demand with matched CME option activity in the smaller matched-option sample. This comparison is included as a mechanism check only. The most interpretable comparison is the D3-plus specification for feeder cattle. In this sample, severe drought does not appear to shift demand from LRP into matched option activity; if anything, both LRP activity and matched option volume decline under D3-plus drought exposure.

Table 15: D4 effects by stronger cattle-exposure measures

Exposure group	Term	County FE	County trends
High cattle + high pre-LRP	Additional effect	-0.039 (0.032)	-0.018 (0.033)
High cattle + high pre-LRP	Base D4 effect	-0.018*** (0.007)	-0.020*** (0.006)
Top decile inventory	Additional effect	-0.021 (0.013)	-0.023* (0.013)
Top decile inventory	Base D4 effect	-0.013* (0.008)	-0.013* (0.007)
Top quartile cattle decline	Additional effect	0.003 (0.016)	0.012 (0.016)
Top quartile cattle decline	Base D4 effect	-0.016** (0.007)	-0.019*** (0.007)
Top quartile inventory	Additional effect	-0.027** (0.013)	-0.021* (0.012)
Top quartile inventory	Base D4 effect	-0.002 (0.010)	-0.006 (0.010)

All models use the balanced feeder cattle county-quarter panel and include county and year-quarter fixed effects. County trend models add county-specific linear trends. Cattle inventory and cattle decline are built from NASS Census county data.

Table 16: D4 drought persistence and feeder LRP demand

D4 exposure	Active quarter	ln policies	ln insured head	ln liability
1 quarter	-0.017*** (0.007)	-0.033*** (0.013)	-0.115*** (0.039)	-0.237*** (0.085)
2 consecutive quarters	-0.023*** (0.008)	-0.037** (0.016)	-0.098* (0.053)	-0.249** (0.111)
3+ consecutive quarters	-0.007 (0.009)	-0.020 (0.018)	-0.054 (0.060)	-0.100 (0.126)

The table compares quarters with one, two, and three or more consecutive quarters of D4 exposure to quarters without current D4 exposure. Models include county and year-quarter fixed effects, with standard errors clustered by county.

Table 17: Lead-lag estimates around first D4 drought onset

Event quarter	Coefficient	N
-4	-0.073 (0.054)	87120
-3	-0.148*** (0.048)	87120
-2	-0.124** (0.049)	87120
0	-0.307*** (0.049)	87120
1	-0.310*** (0.051)	87120
2	-0.260*** (0.049)	87120
3	-0.282*** (0.049)	87120
4	-0.218*** (0.053)	87120

The omitted period is one quarter before the first observed D4 exposure in a county. The sample is the balanced feeder cattle county-quarter panel. All models include county and year-quarter fixed effects, with standard errors clustered by county.

Table 18: Balanced county-quarter drought sample

Cattle type	Counties	County-quarters	Active quarters	Active share	Mean head	Mean endorsements	Mean D4
fed	360	28,880	2,493	8.6%	99.0	0.49	1.79
feeder	1,084	87,120	13,833	15.9%	151.2	1.16	2.06

Notes: Inactive county-quarters are retained with zero LRP activity. Drought exposure is the average county-area percentage.

Table 19: Drought thresholds and insured head

Drought threshold	Cattle type	Estimate	Observations
D1 plus	feeder	0.001 (0.002)	87,120
D2 plus	feeder	0.000 (0.003)	87,120
D3 plus	feeder	-0.006 (0.004)	87,120
D4	feeder	-0.020*** (0.007)	87,120
D1 plus	fed	0.014*** (0.004)	28,880
D2 plus	fed	0.009 (0.006)	28,880
D3 plus	fed	0.008 (0.009)	28,880
D4	fed	0.018 (0.015)	28,880

Notes: The outcome is log one plus insured head. Coefficients represent a 10-percentage-point increase in county drought exposure.

Table 20: Current and lagged extreme drought

Specification	Cattle type	Estimate	Observations
Current D3 plus	feeder	-0.004 (0.004)	84,952
Lag 1 quarter D3 plus	feeder	-0.005 (0.004)	84,952
Lag 2 quarters D3 plus	feeder	0.004 (0.004)	84,952
Current D3 plus	fed	0.016** (0.008)	28,160
Lag 1 quarter D3 plus	fed	-0.016** (0.007)	28,160
Lag 2 quarters D3 plus	fed	0.006 (0.008)	28,160

Notes: The outcome is log one plus insured head. Current and one-quarter-lagged drought exposures are estimated in the balanced panel.

Table 21: D4 effects by pre-period LRP use

Component	Cattle type	Estimate	Observations
Low pre-use counties	feeder	-0.022*** (0.006)	87,120
Additional effect in high pre-use counties	feeder	0.018 (0.034)	87,120
Low pre-use counties	feeder	-0.024*** (0.006)	87,120
Additional effect in high pre-use counties	feeder	0.039 (0.033)	87,120
Low pre-use counties	feeder	-0.008** (0.004)	87,120
Additional effect in high pre-use counties	feeder	0.024 (0.020)	87,120
Low pre-use counties	feeder	-0.013*** (0.004)	87,120
Additional effect in high pre-use counties	feeder	0.032 (0.019)	87,120

Notes: High pre-use counties are defined from feeder LRP use before 2020. The interaction tests whether D4 associations differ in established markets.

Table 22: D4 heterogeneity by cattle and hay exposure

Specification	Component	Estimate	Observations
D4 by high cattle inventory	Base D4 effect	-0.012 (0.008)	86,720
D4 by high cattle inventory	Additional high exposure effect	-0.024* (0.013)	86,720
D4 by high cattle inventory with trends	Base D4 effect	-0.013* (0.007)	86,720
D4 by high cattle inventory with trends	Additional high exposure effect	-0.022* (0.013)	86,720
D4 by high hay dependence	Base D4 effect	-0.018** (0.007)	86,720
D4 by high hay dependence	Additional high exposure effect	-0.018 (0.015)	86,720
D4 by high hay dependence with trends	Base D4 effect	-0.019*** (0.007)	86,720
D4 by high hay dependence with trends	Additional high exposure effect	-0.010 (0.015)	86,720

Notes: County cattle inventory and hay measures come from USDA NASS Census of Agriculture data. Interaction estimates are exploratory.

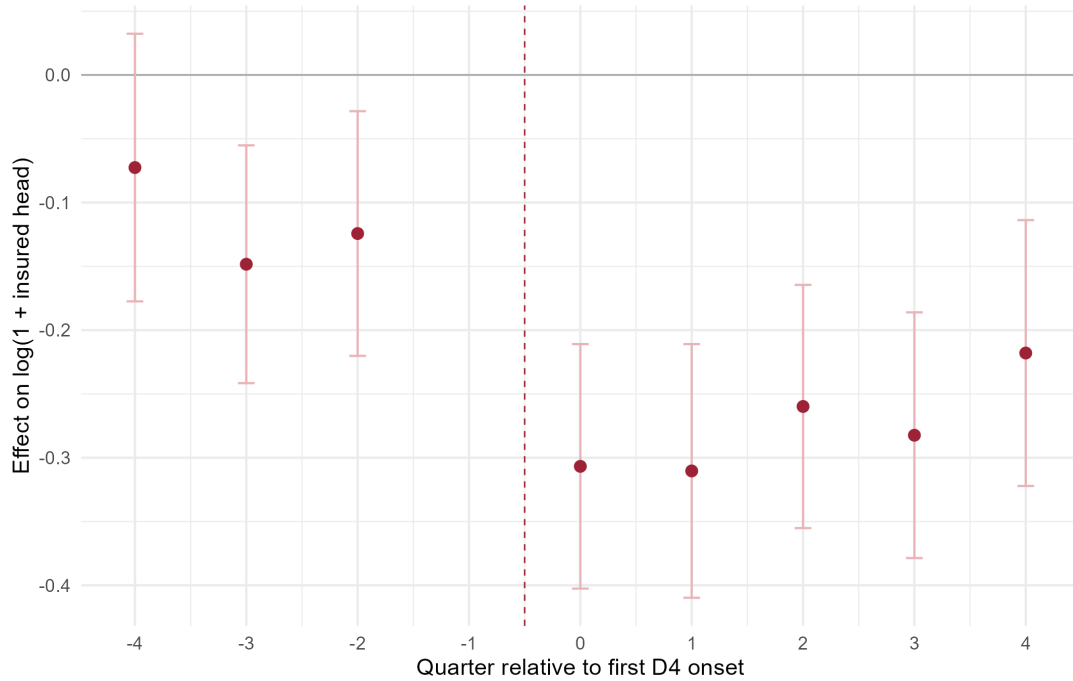


Figure 5: Lead-lag estimates around first D4 drought onset

Table 23: Drought, LRP demand, and corrected matched-option activity

Cattle type	Drought	Outcome	Estimate	Observations
feeder	D3 plus	ln(1+insured head)	-0.009* (0.005)	12,603
feeder	D4	ln(1+insured head)	-0.015 (0.010)	12,603
feeder	D3 plus	ln(1+policies)	-0.009*** (0.003)	12,603
feeder	D4	ln(1+policies)	-0.024*** (0.005)	12,603
feeder	D3 plus	ln(1+liability)	-0.009* (0.005)	12,603
feeder	D4	ln(1+liability)	-0.014 (0.010)	12,603
feeder	D3 plus	ln(1+matched option volume)	-0.007 (0.006)	12,603
feeder	D4	ln(1+matched option volume)	-0.038*** (0.010)	12,603
fed	D3 plus	ln(1+insured head)	0.001 (0.012)	2,169
fed	D4	ln(1+insured head)	0.020 (0.020)	2,169
fed	D3 plus	ln(1+policies)	-0.003 (0.007)	2,169
fed	D4	ln(1+policies)	-0.009 (0.012)	2,169
fed	D3 plus	ln(1+liability)	0.002 (0.012)	2,169
fed	D4	ln(1+liability)	0.021 (0.020)	2,169
fed	D3 plus	ln(1+matched option volume)	-0.016 (0.015)	2,169
fed	D4	ln(1+matched option volume)	-0.037 (0.030)	2,169

Notes: Each row is a separate county and year-quarter fixed-effects regression. Drought is scaled in 10-percentage-point increments. Matched option volume is observed only for LRP endorsements in the corrected option-matched sample.

A.6 Mechanism and heterogeneity

Table 24 reports reduced-form diagnostics for entry-related and coverage outcomes. Several interactions are negative, and the event-study tests show differential pre-reform movement in key outcomes. These estimates do not identify the causal effect of the 2020Q3 reform. They show why the matched-option FEIV design is needed and why producer entry cannot be inferred from endorsement counts alone.

Table 24: Entry and coverage-upgrade diagnostics

Outcome	Sample	Estimate	Observations
Any LRP	balanced extensive	-0.044* (0.025)	23,074
ln(1+head)	balanced extensive	-0.338* (0.183)	23,074
ln(1+policies)	balanced extensive	-0.170** (0.073)	23,074
ln(1+liability)	balanced extensive	-0.654* (0.349)	23,074
ln average coverage	conditional positive upgrade	-0.014*** (0.001)	8,800
Share $\geq 97\%$	conditional positive upgrade	-0.189*** (0.019)	8,800
Average coverage	conditional positive upgrade	-0.013*** (0.001)	8,800

Notes: The coefficient is the interaction between the 2020Q3-or-later indicator and a predetermined high-coverage-use indicator. Significant pre-reform trends prevent a causal difference-in-differences interpretation.

Table 25 reports results by cattle type, pre-reform use, and region. The feeder high-tier result is most precisely estimated in the Plains, where LRP activity is concentrated. Other regional and pre-use estimates have wider confidence intervals. These subgroup models

are exploratory.

Table 25: FEIV heterogeneity by cattle type, pre-use, and region

Dimension	Group	Outcome	Estimate	Excluded-instrument F
cattle type	feeder	ln insured head	0.197* (0.108)	1089.3
cattle type	feeder	ln endorsements	0.505*** (0.060)	1089.3
cattle type	feeder	ln liability	0.189* (0.111)	1089.3
cattle type	feeder	ln average coverage	0.003* (0.002)	1089.3
cattle type	feeder	Share $\geq 97\%$	0.208*** (0.027)	1089.3
cattle type	fed	ln insured head	0.312 (0.419)	182.3
cattle type	fed	ln endorsements	1.123*** (0.265)	182.3
cattle type	fed	ln liability	0.307 (0.422)	182.3
cattle type	fed	ln average coverage	-0.010*** (0.004)	182.3
cattle type	fed	Share $\geq 97\%$	0.126 (0.097)	182.3
pre use size	zero pre use	ln insured head	0.339* (0.199)	515.6
pre use size	zero pre use	ln average coverage	0.009*** (0.003)	515.6
pre use size	zero pre use	Share $\geq 97\%$	0.228*** (0.044)	515.6
pre use size	low pre use	ln insured head	-0.197 (0.276)	186.1
pre use size	low pre use	ln average coverage	0.004 (0.003)	186.1
pre use size	low pre use	Share $\geq 97\%$	0.161** (0.080)	186.1
pre use size	mid pre use	ln insured head	0.320 (0.219)	205.1
pre use size	mid pre use	ln average coverage	0.004 (0.004)	205.1
pre use size	mid pre use	Share $\geq 97\%$	0.239*** (0.061)	205.1
pre use size	high pre use	ln insured head	0.234 (0.187)	421.2
pre use size	high pre use	ln average coverage	0.000 (0.003)	421.2
pre use size	high pre use	Share $\geq 97\%$	0.223*** (0.044)	421.2
region	Midwest	ln insured head	-0.211 (0.304)	266.0
region	Midwest	ln average coverage	0.007** (0.003)	266.0
region	Midwest	Share $\geq 97\%$	0.137* (0.082)	266.0
region	Plains	ln insured head	0.168 (0.118)	833.0
region	Plains	ln average coverage	0.003 (0.002)	833.0
region	Plains	Share $\geq 97\%$	0.213*** (0.030)	833.0
region	South	ln insured head	0.761* (0.458)	122.9
region	South	ln average coverage	0.005 (0.005)	122.9
region	South	Share $\geq 97\%$	0.029 (0.102)	122.9
region	WestOther	ln insured head	1.614* (0.826)	66.7
region	WestOther	ln average coverage	0.004 (0.006)	66.7
region	WestOther	Share $\geq 97\%$	0.172* (0.103)	66.7

Notes: Models use the corrected main matching rule and county-clustered standard errors. Subgroup estimates are exploratory and are not formal tests that coefficients are equal across groups.

Table 26 reports leave-one-top-state-out estimates. This check addresses the concern that the main coverage result is driven by a single high-use state. The coverage estimates remain positive when major states are excluded one at a time.

A.7 Counterfactual policy calculations

Tables 28 and 29 report calculations for policies effective from 2020Q3 through 2024. The exercise replaces observed subsidies with a flat 20 percent premium subsidy and applies

Table 26: Leave-one-top-state-out FEIV estimates

Excluded state	Outcome	Estimate	Observations	Excluded-instrument F
SD	ln insured head	0.142 (0.117)	10,760	910.7
SD	ln average coverage	0.002 (0.002)	10,760	910.7
SD	Share $\geq 97\%$	0.215*** (0.029)	10,760	910.7
NE	ln insured head	0.272** (0.117)	10,757	874.9
NE	ln average coverage	0.003 (0.002)	10,757	874.9
NE	Share $\geq 97\%$	0.218*** (0.029)	10,757	874.9
TX	ln insured head	0.220* (0.114)	11,733	1033.3
TX	ln average coverage	0.002 (0.002)	11,733	1033.3
TX	Share $\geq 97\%$	0.209*** (0.028)	11,733	1033.3
OK	ln insured head	0.167 (0.113)	11,572	974.2
OK	ln average coverage	0.004*** (0.002)	11,572	974.2
OK	Share $\geq 97\%$	0.190*** (0.027)	11,572	974.2
KS	ln insured head	0.157 (0.118)	10,665	953.3
KS	ln average coverage	0.004** (0.002)	10,665	953.3
KS	Share $\geq 97\%$	0.207*** (0.032)	10,665	953.3
IA	ln insured head	0.211* (0.110)	11,915	1033.3
IA	ln average coverage	0.003 (0.002)	11,915	1033.3
IA	Share $\geq 97\%$	0.207*** (0.028)	11,915	1033.3
ND	ln insured head	0.247** (0.114)	11,566	1079.1
ND	ln average coverage	0.002 (0.002)	11,566	1079.1
ND	Share $\geq 97\%$	0.208*** (0.029)	11,566	1079.1
MO	ln insured head	0.210* (0.111)	11,817	1027.9
MO	ln average coverage	0.003 (0.002)	11,817	1027.9
MO	Share $\geq 97\%$	0.210*** (0.028)	11,817	1027.9
CO	ln insured head	0.209* (0.109)	12,234	1048.9
CO	ln average coverage	0.003 (0.002)	12,234	1048.9
CO	Share $\geq 97\%$	0.201*** (0.028)	12,234	1048.9
MT	ln insured head	0.182* (0.108)	12,237	1073.2
MT	ln average coverage	0.003* (0.002)	12,237	1073.2
MT	Share $\geq 97\%$	0.209*** (0.027)	12,237	1073.2
NM	ln insured head	0.199* (0.108)	12,453	1079.9
NM	ln average coverage	0.003* (0.002)	12,453	1079.9
NM	Share $\geq 97\%$	0.209*** (0.027)	12,453	1079.9
CA	ln insured head	0.181* (0.107)	12,488	1082.7
CA	ln average coverage	0.003* (0.002)	12,488	1082.7
CA	Share $\geq 97\%$	0.209*** (0.027)	12,488	1082.7

Notes: Each model removes one of the twelve largest feeder-LRP states by full-panel insured head and reestimates the corrected main FEIV specification.

Table 27: State-level feeder-cattle FEIV estimates

State	Outcome	Estimate	Observations	Counties	Excluded-instrument F
CO	ln insured head	0.125 (0.981)	316	30	30.9
CO	ln average coverage	0.003 (0.009)	316	30	30.9
CO	Share $\geq 97\%$	0.468*** (0.141)	316	30	30.9
IA	ln insured head	-0.190 (0.599)	650	79	111.6
IA	ln average coverage	0.006 (0.006)	650	79	111.6
IA	Share $\geq 97\%$	0.136 (0.153)	650	79	111.6
KS	ln insured head	0.399* (0.229)	1,913	101	196.8
KS	ln average coverage	-0.002 (0.004)	1,913	101	196.8
KS	Share $\geq 97\%$	0.221*** (0.055)	1,913	101	196.8
MO	ln insured head	-0.224 (0.371)	747	69	154.5
MO	ln average coverage	0.007 (0.005)	747	69	154.5
MO	Share $\geq 97\%$	0.182* (0.104)	747	69	154.5
MT	ln insured head	1.077 (1.823)	327	35	7.9
MT	ln average coverage	0.009 (0.013)	327	35	7.9
MT	Share $\geq 97\%$	0.042 (0.212)	327	35	7.9
ND	ln insured head	-0.009 (0.337)	1,010	51	81.0
ND	ln average coverage	0.007* (0.004)	1,010	51	81.0
ND	Share $\geq 97\%$	0.167** (0.074)	1,010	51	81.0
NE	ln insured head	-0.107 (0.301)	1,821	85	253.4
NE	ln average coverage	0.005 (0.005)	1,821	85	253.4
NE	Share $\geq 97\%$	0.186** (0.073)	1,821	85	253.4
OK	ln insured head	0.409 (0.382)	1,003	53	209.8
OK	ln average coverage	-0.002 (0.005)	1,003	53	209.8
OK	Share $\geq 97\%$	0.330*** (0.085)	1,003	53	209.8
SD	ln insured head	0.727*** (0.257)	1,818	66	204.7
SD	ln average coverage	0.006 (0.004)	1,818	66	204.7
SD	Share $\geq 97\%$	0.147* (0.081)	1,818	66	204.7
TX	ln insured head	0.044 (0.303)	826	111	72.9
TX	ln average coverage	0.015*** (0.005)	826	111	72.9
TX	Share $\geq 97\%$	0.195* (0.104)	826	111	72.9

Notes: States enter when the matched sample contains at least 300 observations and ten counties. Estimates are exploratory.

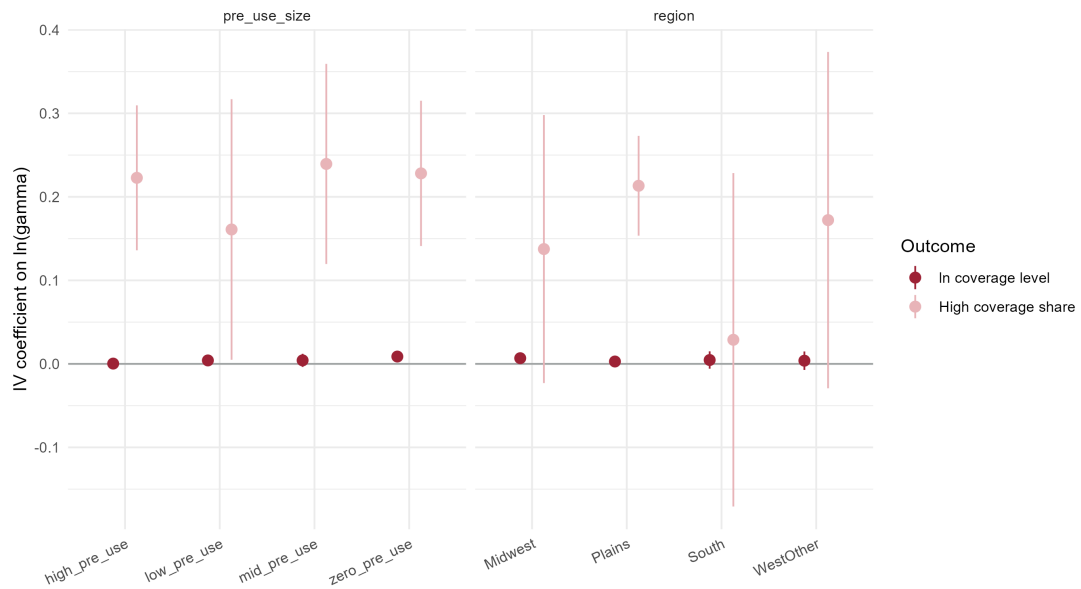


Figure 6: Coverage response by pre-reform use and region

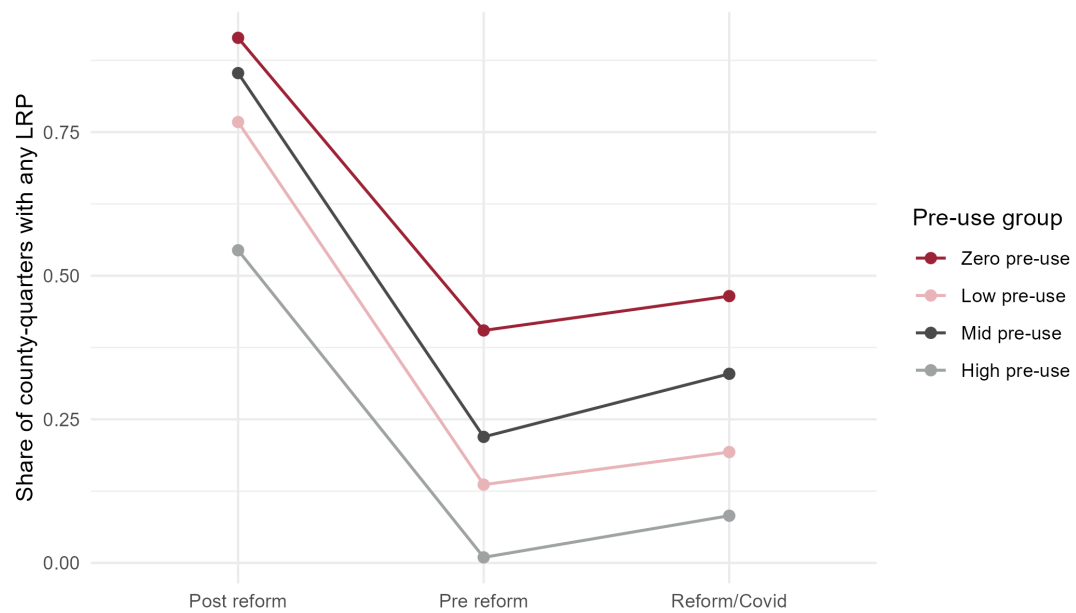


Figure 7: Entry margin patterns by pre-reform LRP use

the corrected main high-tier coefficient. It summarizes fiscal cost under the maintained model and is not a welfare analysis.

Table 28: Counterfactual cost of the tiered subsidy schedule

Period	Insured head	Extra subsidy	Predicted high-tier head	Cost per predicted head
2020 Q3–2021	1,379,728	\$ 10,757,221	50,381	\$214
2022–2024	10,983,621	\$125,031,110	439,792	\$284
2020 Q3–2024 total	12,363,349	\$135,788,331	490,172	\$277

Notes: The counterfactual begins in 2020Q3 and ends in 2024. It replaces observed subsidy payments with a flat 20 percent premium subsidy and applies the corrected main high-tier FEIV coefficient. Predicted share changes are capped at the available head share. This is not a welfare estimate.

Table 29: Counterfactual cost by pre-reform use

Pre-use group	Insured head	Extra subsidy	Predicted high-tier head	Cost per predicted head
zero pre use	6,552,363	\$71,242,391	228,759	\$311
high pre use	2,503,381	\$28,646,232	95,540	\$300
low pre use	1,458,784	\$16,145,929	58,138	\$278
mid pre use	1,848,821	\$19,753,779	107,735	\$183

Notes: Pre-use groups are defined from 2015–2018 feeder-LRP insured head. Calculations use the capped predicted high-tier response over 2020Q3–2024.

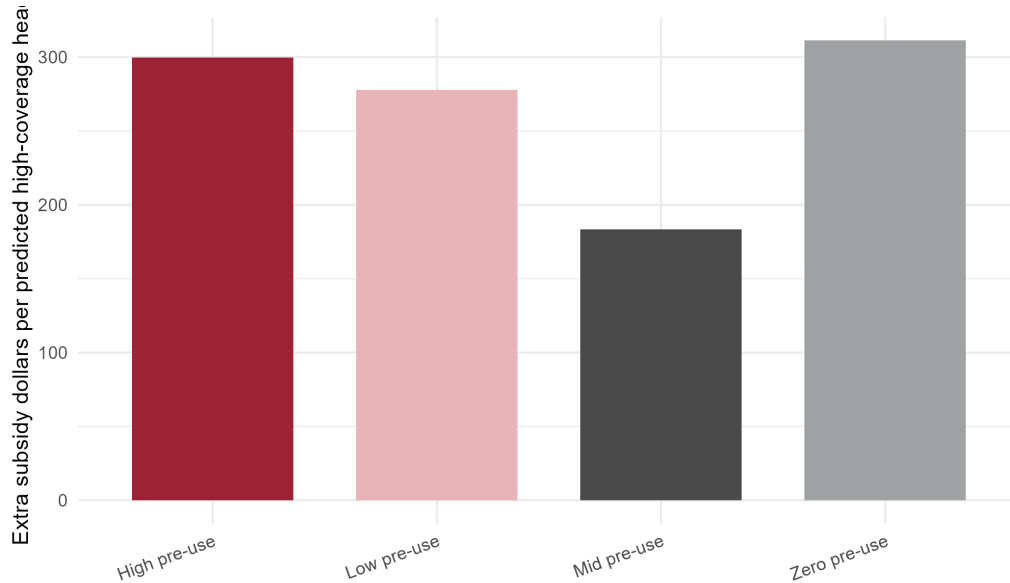


Figure 8: Counterfactual cost per predicted high-coverage head by pre-reform use

A.8 Spatial concentration

Tables 30 and 31 report the largest feeder-cattle LRP states and counties by insured head. The calculations use the independent full-LRP panel and policy effective dates observed through July 2025.

Table 30: Top feeder-cattle LRP states by insured head

State	Insured head	National share	Endorsements	High-tier head share
SD	2,520,296	14.5%	24,848	91.6%
NE	2,237,758	12.9%	17,304	91.7%
TX	2,221,444	12.8%	10,821	91.5%
OK	1,854,234	10.7%	13,852	79.1%
KS	1,689,544	9.7%	17,441	75.1%
IA	1,054,400	6.1%	8,317	94.2%
ND	870,714	5.0%	9,227	83.3%
MO	673,940	3.9%	7,892	75.5%
CO	626,762	3.6%	3,656	93.9%
MT	511,911	2.9%	3,029	93.4%

Notes: Totals use the independent full-LRP panel and policy effective dates from January 2005 through July 24, 2025.

Table 31: Top feeder-cattle LRP counties by insured head

State	County FIPS	Insured head	National share	High-tier head share
IA	19167	421,020	2.42%	99.1%
NM	35009	309,211	1.78%	98.4%
TX	48117	248,630	1.43%	91.3%
SD	46115	206,290	1.19%	92.2%
SD	46005	201,285	1.16%	90.7%
TX	48369	178,775	1.03%	99.0%
CO	08063	167,985	0.97%	97.1%
SD	46059	165,519	0.95%	95.7%
OK	40003	163,943	0.94%	85.0%
MO	29109	160,946	0.93%	78.3%

Notes: County totals use the independent full-LRP panel and policy effective dates from January 2005 through July 24, 2025.

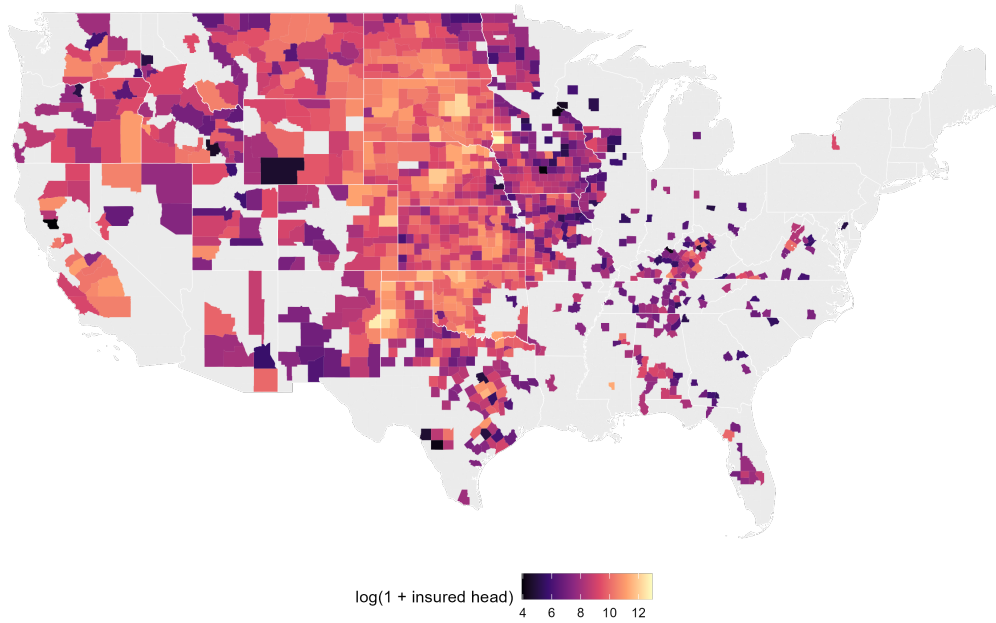


Figure 9: County-level feeder cattle LRP insured head

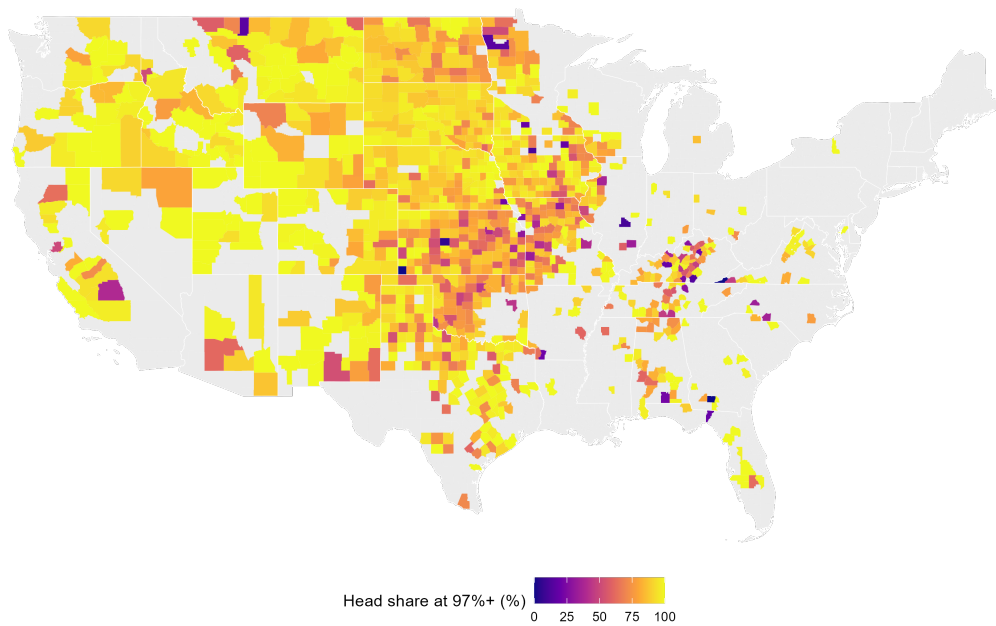


Figure 10: County-level high-coverage share in feeder cattle LRP